

first light

digitization and programming · setup and first steps · prof. dr. nicolas meseth

- 1 your device
- 2 your tools
- 3 getting ready
- 4 how the workshop works

meet
your device.



image: ai-generated from photos of the real device (gpt-image-2.5-sunburst)

part 1

your device

your device

the eye

a colour sensor, on the left



your device

your device

the eye

a colour sensor, on the left

the lamp

an RGB LED, on the right



your device

the eye

a colour sensor, on the left

the lamp

an RGB LED, on the right

the wall

so the device does not see its own lamp



your device

your device

the eye

a colour sensor, on the left

the lamp

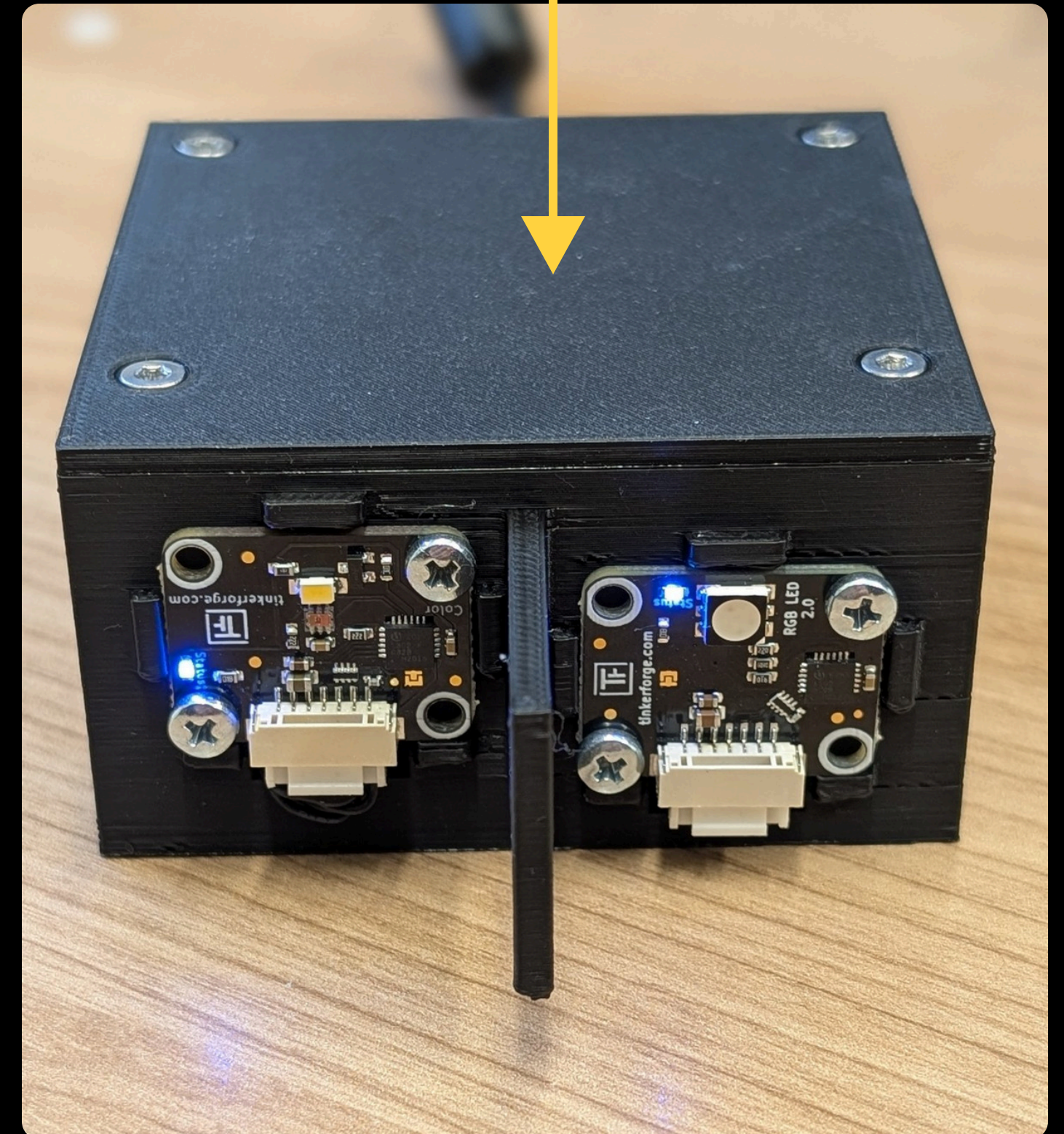
an RGB LED, on the right

the wall

so the device does not see its own lamp

inside

a Master Brick, connected to your laptop by USB



 it's yours until january

two devices per team, one for each of you.
sign for it when you pick it up.
bring it every monday. keep it in one piece.

part 2

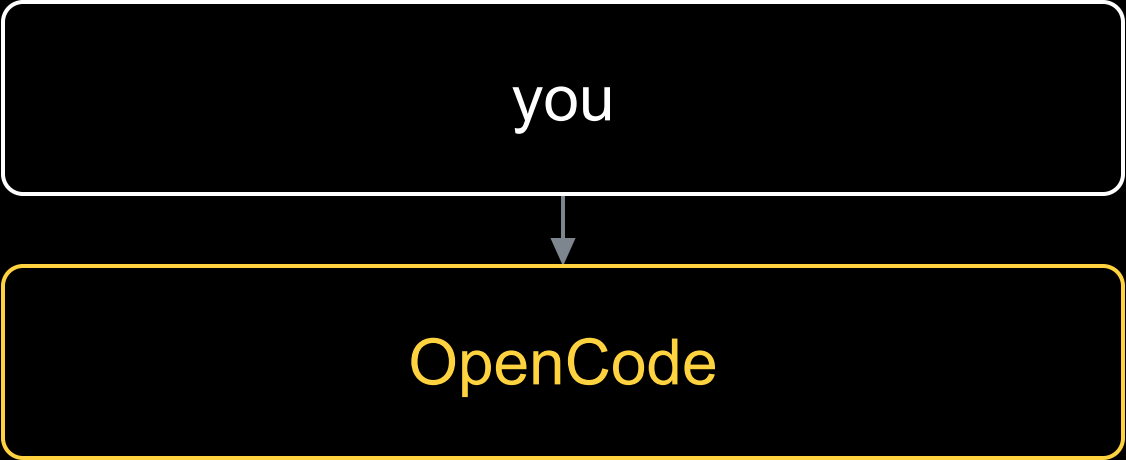
your tools

from your idea to the light



decide what should happen

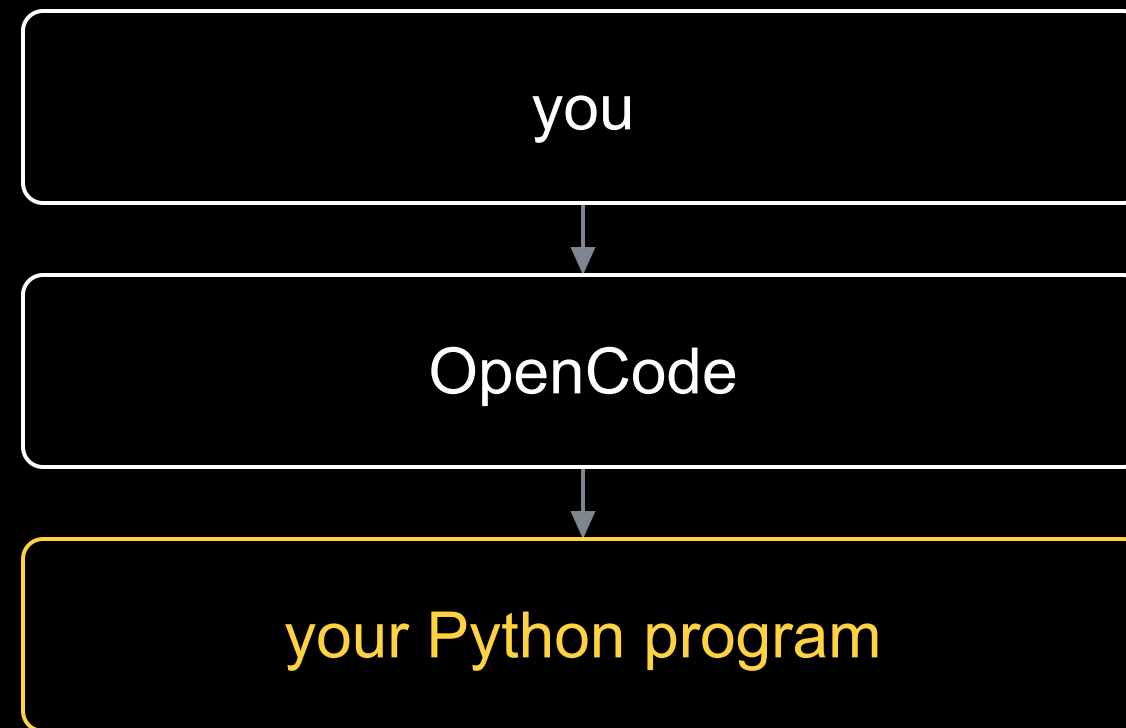
from your idea to the light



decide what should happen

your assistant writes and explains the program

from your idea to the light

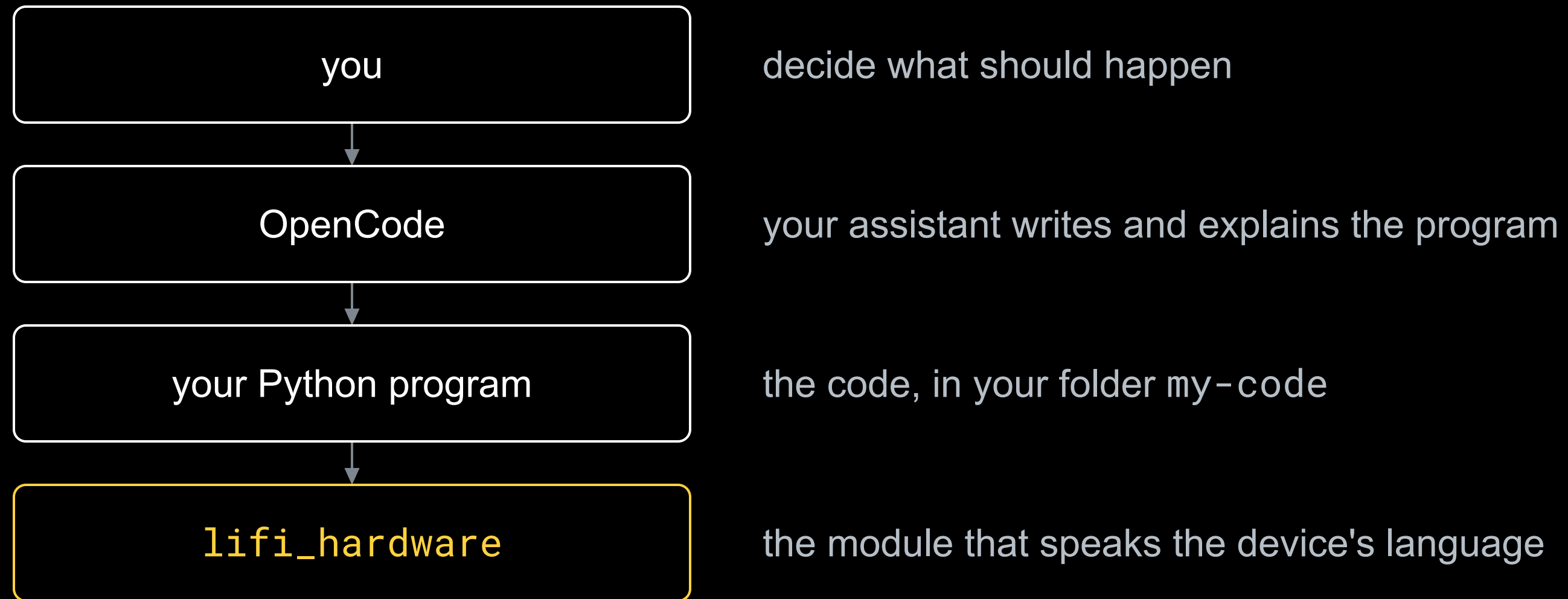


decide what should happen

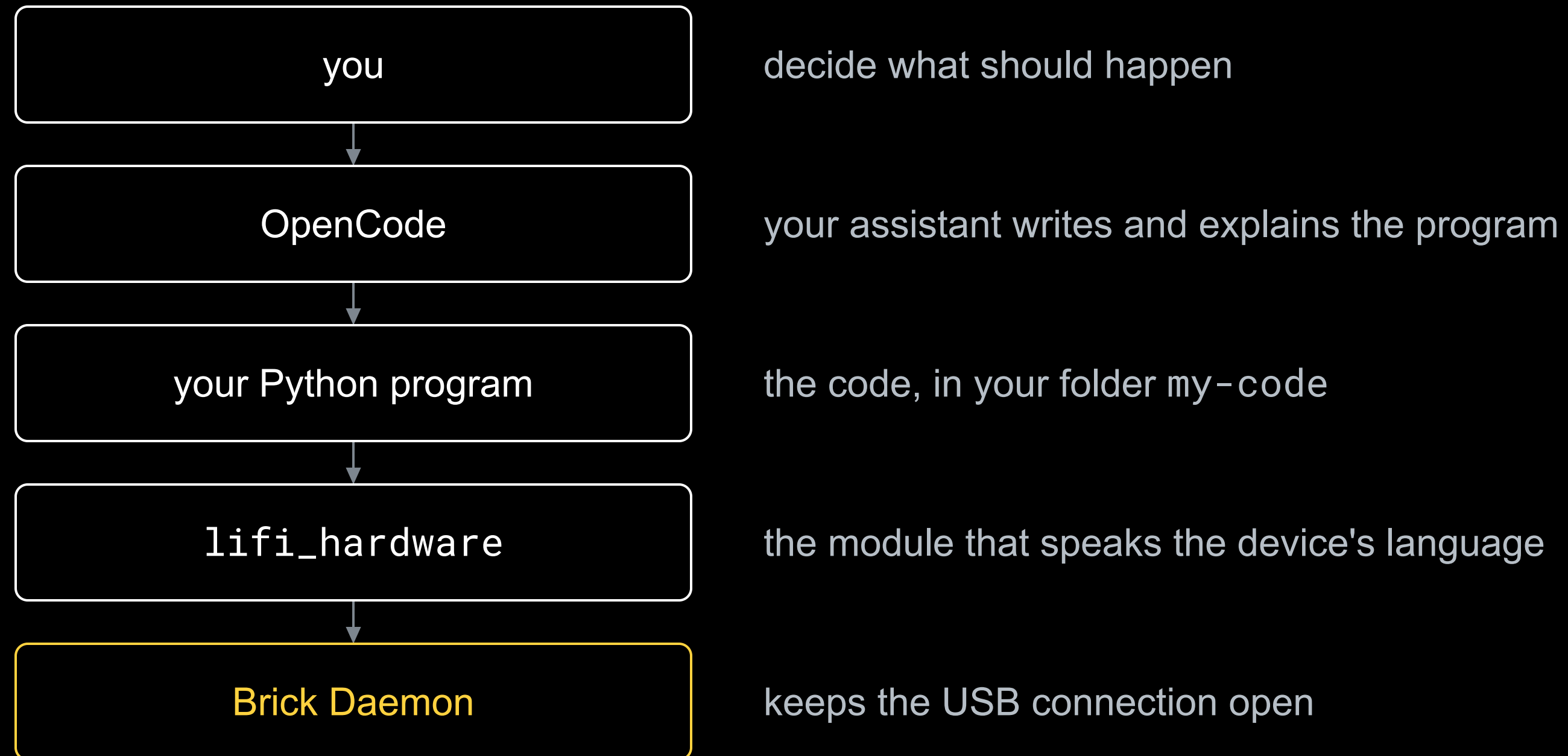
your assistant writes and explains the program

the code, in your folder my-code

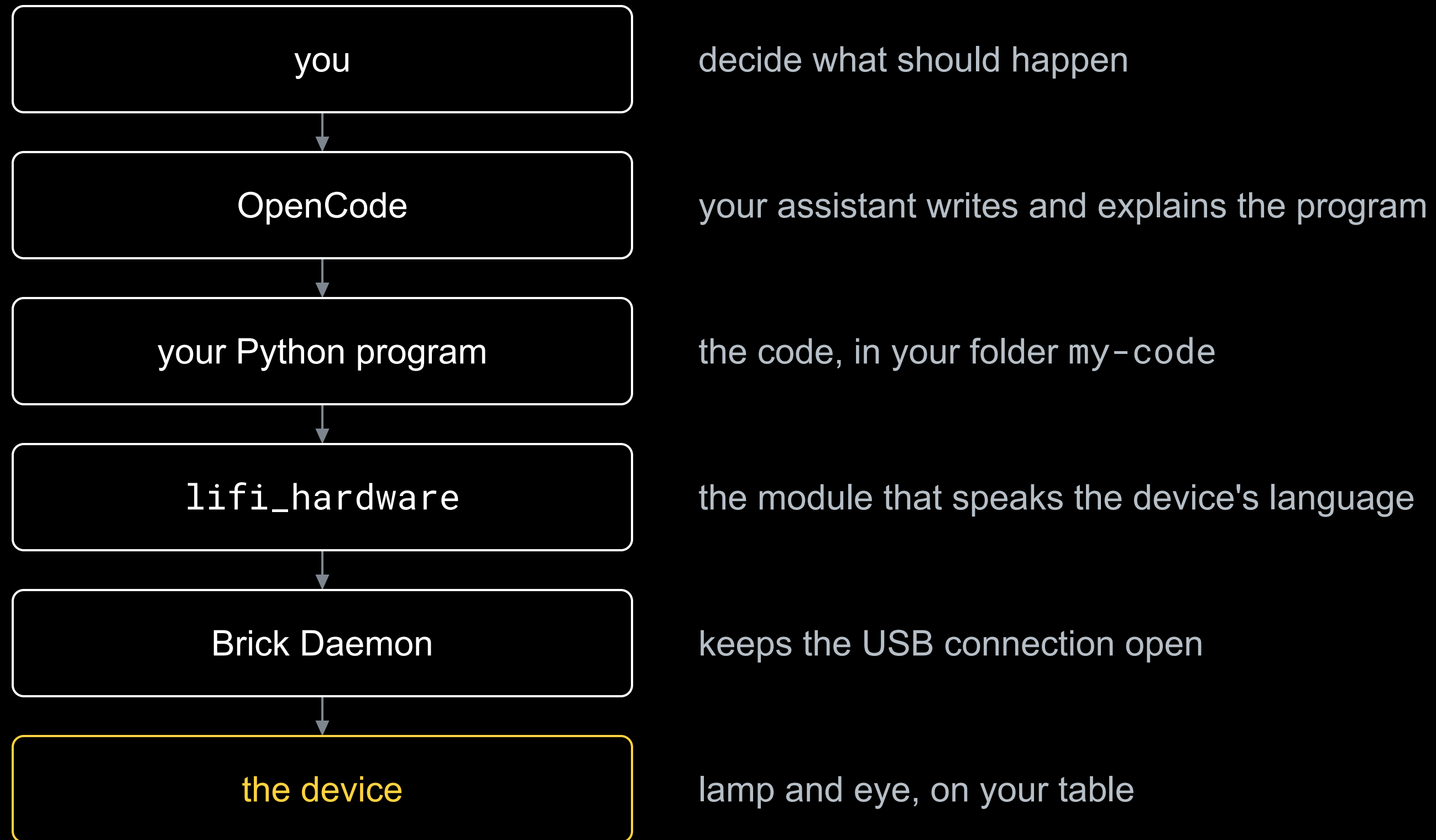
from your idea to the light



from your idea to the light



from your idea to the light



opencode

your assistant, with a model already
set up for this course.



knows the course

the website, the lecture
notes, the module
lifi_hardware.



ask anything

as often as you like. it
explains, installs and writes
code.



ready to go

the model is set up, the
course is loaded. just start
asking.



it cannot see your device

the lamp has the last word.



free models first

OpenCode comes with free models. they work right away, without a key.



a key, if needed

for a stronger model, a key may follow later. you will hear about it.



a key is only for OpenCode

never into a chat, a file or a screenshot. whoever has a key works at its owner's expense.

part 3

getting ready

part 1, by hand

1 install OpenCode Desktop.

step by step: docs.lifi-project.de/software

part 1, by hand

- 1 install OpenCode Desktop.
- 2 download the course folder as a zip and unpack it.

step by step: docs.lifi-project.de/software

part 1, by hand

- 1 install OpenCode Desktop.
- 2 download the course folder as a zip and unpack it.
- 3 open the folder in OpenCode and start a session.

step by step: docs.lifi-project.de/software

part 2, your assistant takes over

/onboarding



image: ai-generated (gpt-image-2.5-flare)

getting ready

part 2, your assistant takes over

/onboarding

it asks you three questions.



image: ai-generated (gpt-image-2.5-flare)

getting ready

part 2, your assistant takes over

/onboarding

it asks you three questions.

it installs Python, lifi_hardware, VS Code, Brick Daemon and Brick Viewer.



image: ai-generated (gpt-image-2.5-flare)

getting ready

part 2, your assistant takes over

/onboarding

it asks you three questions.

it installs Python, lifi_hardware, VS Code, Brick Daemon and Brick Viewer.

it checks every step before the next one.



part 2, your assistant takes over

/onboarding

it asks you three questions.

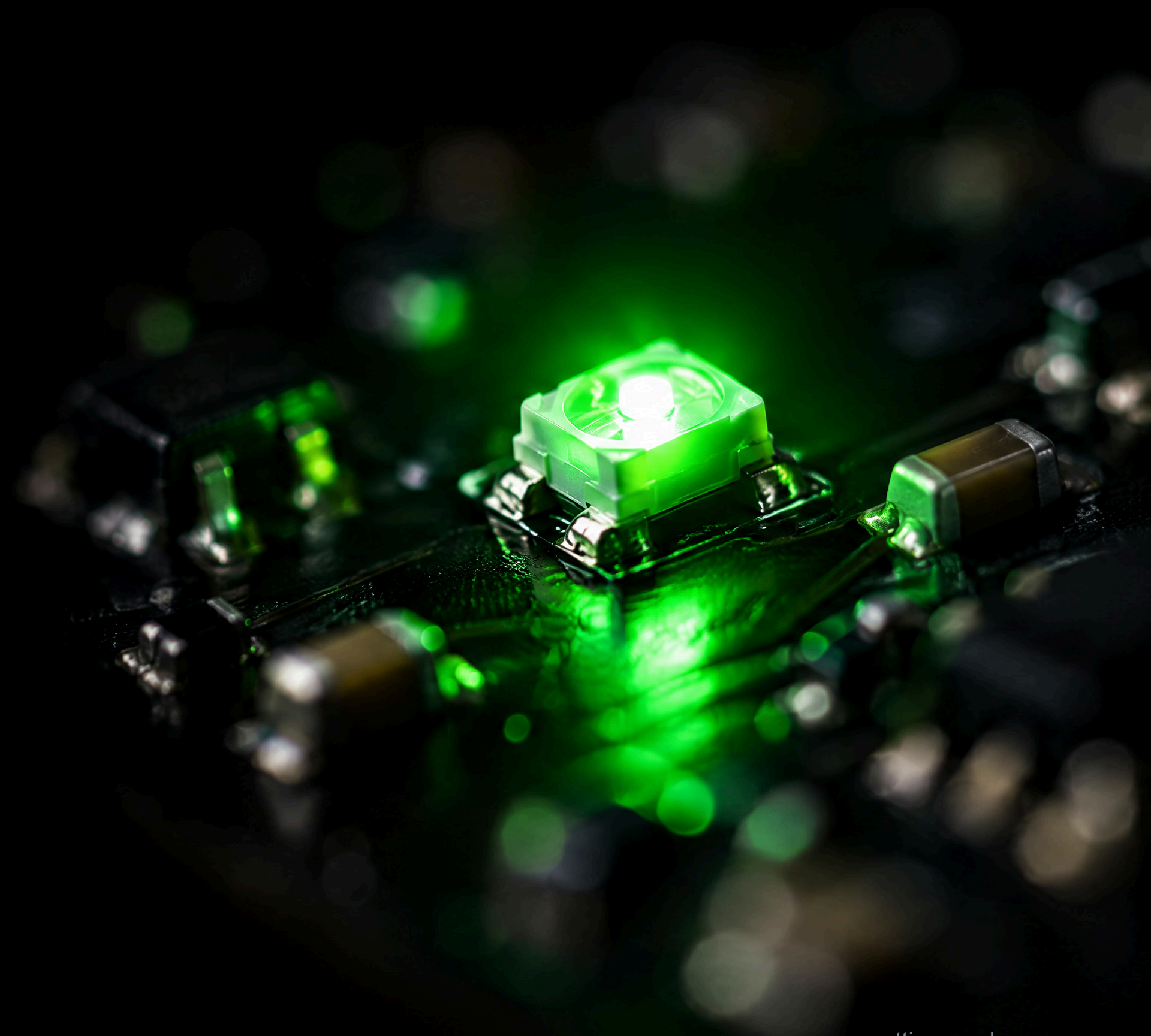
it installs Python, `lifi_hardware`, VS Code, Brick Daemon and Brick Viewer.

it checks every step before the next one.

when it asks, plug in your device.

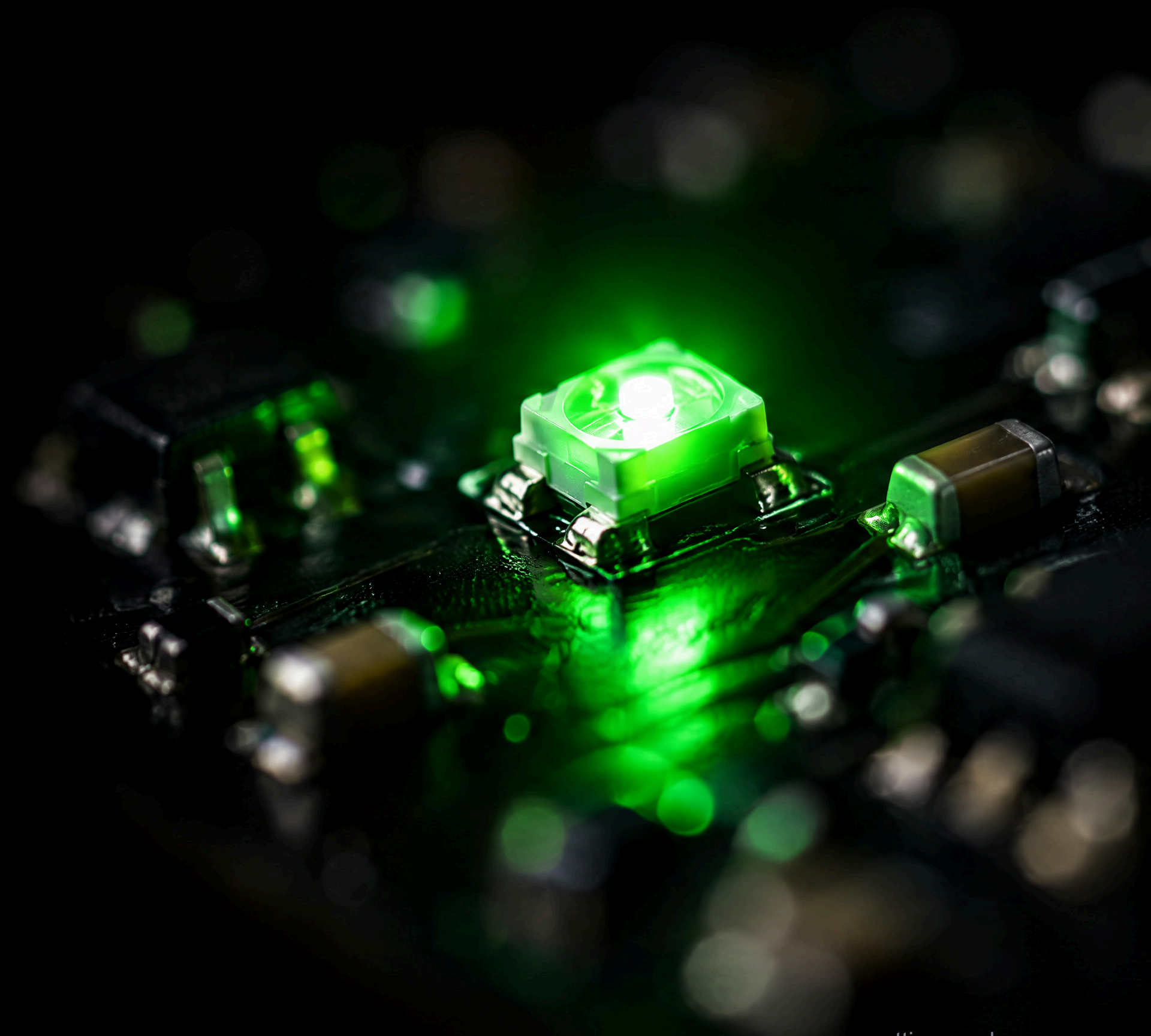


your LED lights up
green.



your LED lights up
green.

and your device appears in the cockpit.



where are you?

- ☐ OpenCode installed
- ☐ course folder open
- ☐ assistant answers
- ☐ /onboarding started
- ☐ LED green
- ☐ device in the cockpit

part 4

how the workshop works

⚙️ stuck? ask in this order

1 your assistant

🌀 stuck? ask in this order

- 1 your assistant
- 2 your partner

🌀 stuck? ask in this order

- 1 your assistant
- 2 your partner
- 3 the team next to you

🎮 stuck? ask in this order

- 1 your assistant
- 2 your partner
- 3 the team next to you
- 4 nicolas

how to ask your assistant

say what you wanted to happen.

paste what happened instead: the whole error, even screenshots.

ask why, not only how.

↻ every week

type `/update-semester`: new material arrives in your folder.
your own work lives in my-code. everything else gets replaced.

until next monday:
finish the setup with /onboarding.
read “the project” and “challenge 0”.

